

TEAM Code and Chaos

CRIME FIR

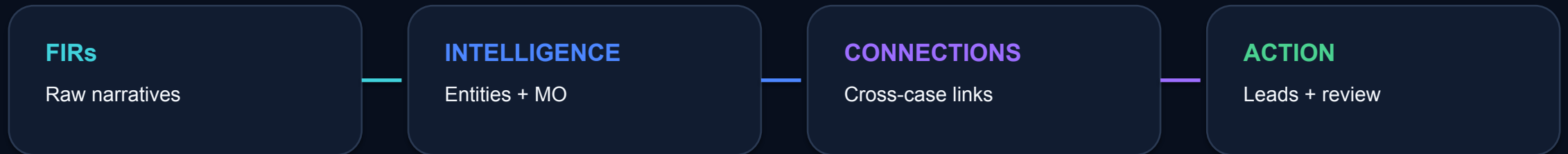
FIR Intelligence & Crime Pattern Detector

From isolated FIR narratives → connected, explainable investigation intelligence.

PS10 • TRACK 4

IBM x NFSU

THE STORY OF THE SYSTEM



Core principle: AI accelerates analysis; investigators remain the decision-makers.

The Problem exists where...

WHAT THE INVESTIGATOR HAS

FIR TEXT

Details are spread across names, places, dates, modus operandi and evidence identifiers.

- Accused / aliases
- Location + station
- MO + victim profile
- Phone / account / device / vehicle

WHAT GETS LOST

CASE CONTEXT

One FIR is readable; the batch is harder to compare.

REPEATED SIGNALS

The same infrastructure or MO may appear across cases.

CROSS-STATION LINKS

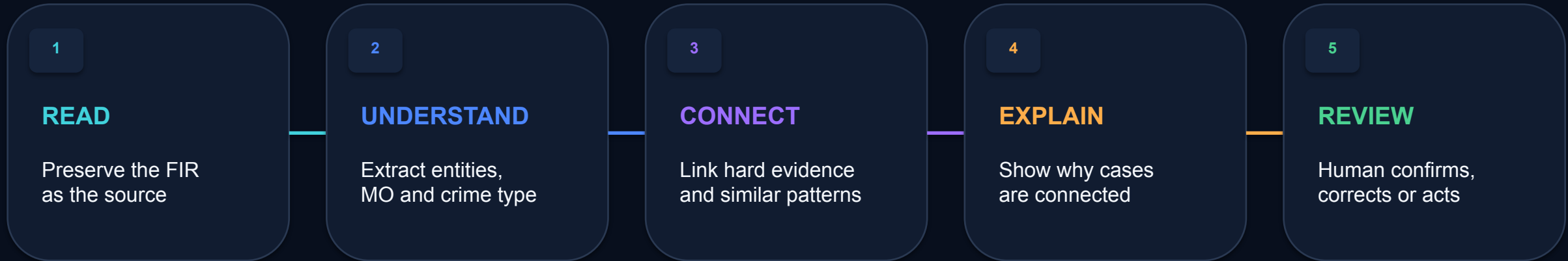
Connections can span stations and districts.

PS10 DEMANDS

- 1 Classify crime type
- 2 Extract entities + MO
- 3 Detect repeat-offender signatures
- 4 Summarize station-level trends

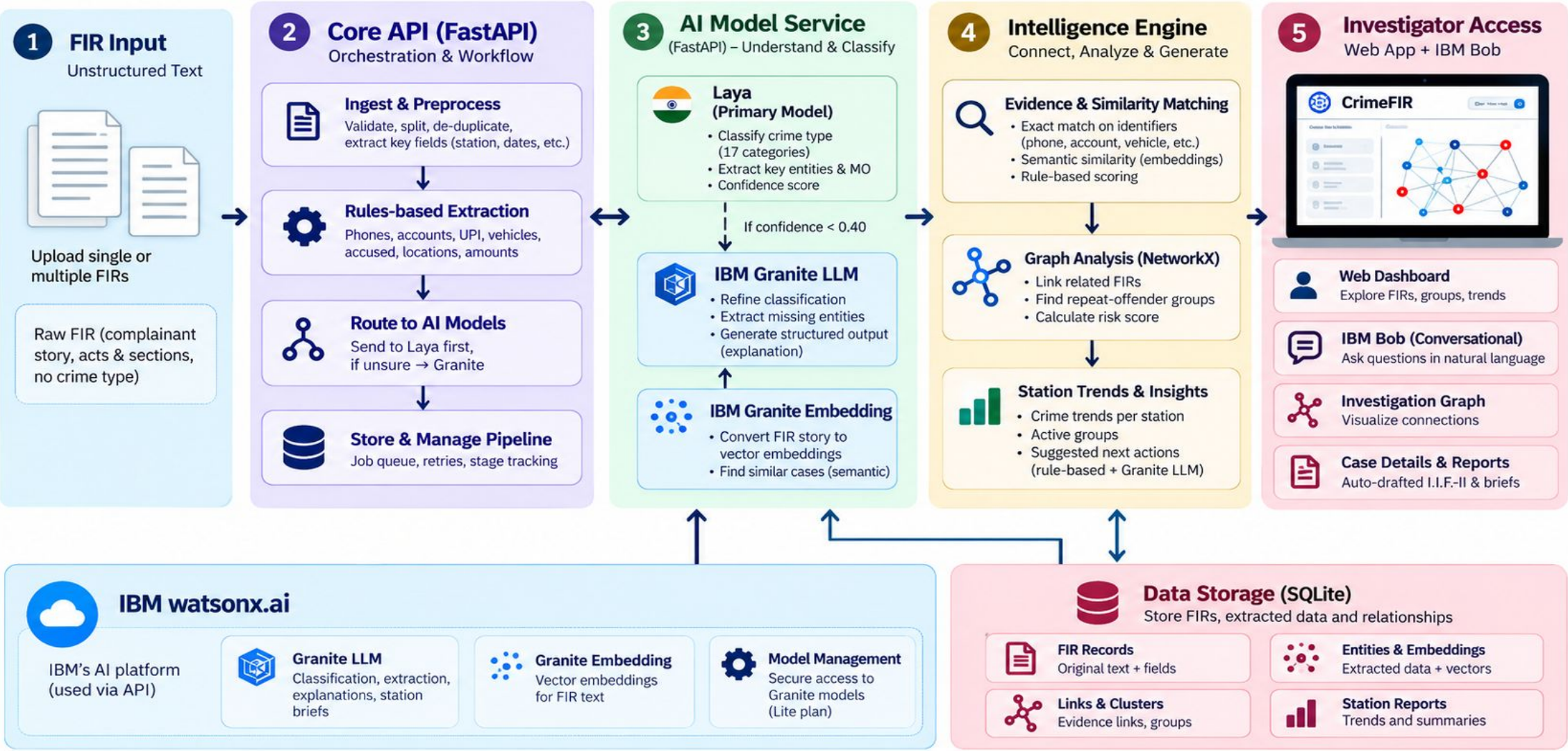
So the design question becomes: How do we turn each FIR into evidence that can be compared, connected and explained?

CrimeFIR creates a chain from text → intelligence → connections → investigation leads.

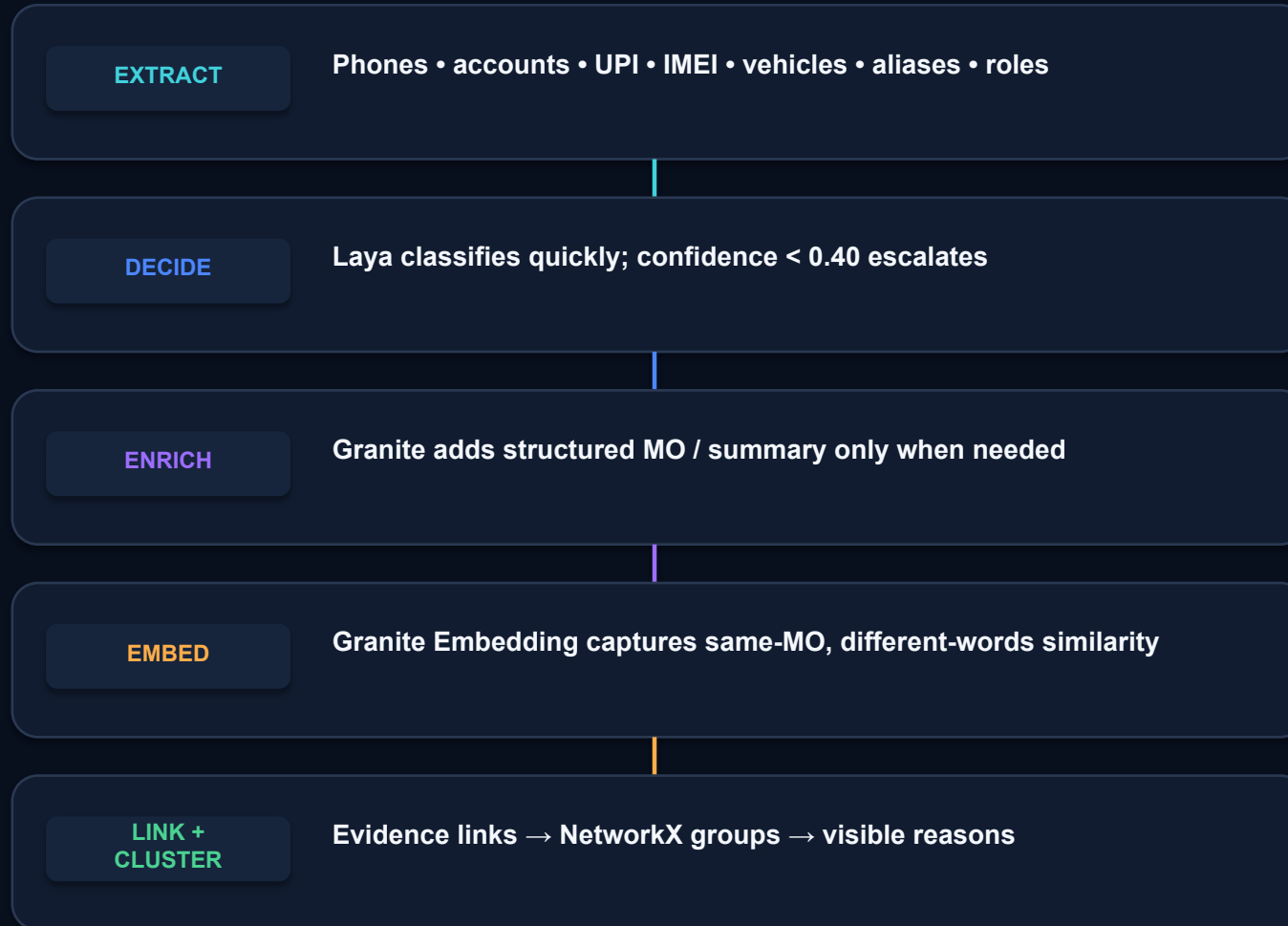


ONE PIPELINE → FIVE USEFUL OUTPUTS





Every FIR is processed as...



TWO TYPES OF CONNECTIONS

EVIDENCE

Shared hard identifier or resolved accused.

→ Stronger, explainable connection

PATTERN

Similar crime story / MO.

→ Investigation lead, not proof

WHAT THE OFFICER SEES

- Why are these FIRs connected?
- Which evidence created the link?
- What should be checked next?

Tech Stack Used

- **Frontend: NextJs, Tailwind CSS**
- **Backend: Python**
- **AI Tools: Laya, IBM Granite, IBM BOB**

IBM technologies are tied to specific jobs in the workflow.

We use IBM where it adds reasoning, semantic understanding or investigator interaction

IBM BOB

INVESTIGATOR INTERFACE

Natural-language questions
MCP tool access
FIR Analyst custom mode

GRANITE LLM

UNCERTAINTY + BRIEFS

Escalated FIR analysis
Structured enrichment
Grounded station briefs

GRANITE EMBEDDING

SEMANTIC MATCHING

Same MO, different words
Pattern similarity
Lead generation

watsonx.ai

Hosts the Granite LLM path used for escalated reasoning and grounded report generation.

Impact of CRIME FIR

Success means investigators can move from raw FIR batches to explainable cross-case intelligence with less manual comparison.

BEFORE CRIME FIR

- FIRs remain case-by-case
- Cross-case comparison is manual
- Patterns are harder to explain
- Station summaries take effort

WITH CRIME FIR

- Every FIR becomes structured intelligence
- Evidence links expose recurring infrastructure
- Graphs show why cases connect
- Station trends become repeatable outputs

SUCCESS LOOKS LIKE

93.4%

Major crime head

unseen test FIRs

83.0%

Exact crime type

unseen test FIRs

100%

Evidence precision

hard identifiers

11 / 12

Planted gangs

exact clusters

0 / 40

Decoys

flagged as gangs

Team & Roles

Member 1: Patel Raj - Team Lead & Architecture

Member 2: Karan yadav - backend & AI

Member 3: Dhaval Patil - Frontend

Member 4: Parth shete - Dataset, Testing & Docs